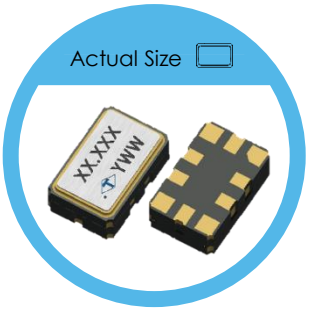
TAITIEN
Our Performance, Your Reputation

TL Type for Stratum 3

5.0 x 3.2 mm SMD Voltage Controlled Temperature Compensated Crystal Oscillator



RoHS Compliant

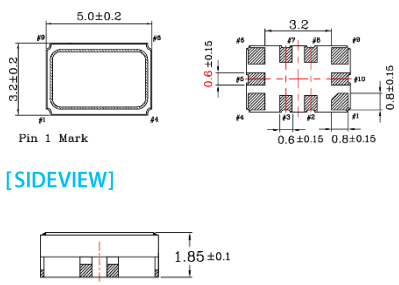
FEATURES

- Typical 5.0 x 3.2 x 1.85 mm ceramic SMD package.
- Stratum 3 (Overall ±4.6ppm including 20 years aging.)
- CMOS and Clipped Sine wave (without DC-cut capacitor) output optional.

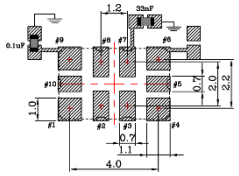
TYPICAL APPLICATION

- Stratum 3
- Femtocell, Base Stations

DIMENSION (mm)



SOLDER PAD LAYOUT (mm)



Recommended Soldering Pattern
To ensure optimal oscillator performance, place a by-pass capacitor of 0.1µF as close to the part as possible between V_{DD} and GND Pads.

PIN FUNCTION (mm)

PIN#	FUNCTION
1	V_{CON}:VC-TCXO GND/NC:TCXO
2	Do not Connect
3	Do not Connect
4	GND
5	Tri-State
6	Fout
7	VC Filter
8	NC
9	V_{DD}
10	GND

ELECTRICAL SPECIFICATION

Parameter			2.5V / 3.3V Min.	Typ.	Max.	Unit	Test Condition
Supply Voltage Variation (V_{DD})			V_{DD} – 5%		V_{DD} + 5%	V	
Frequency Range			10		52	MHz	
Standard Frequency			10, 12.8, 16.384, 19.2, 20, 25, 26			MHz	Standard frequencies are frequencies which the crystal has been designed and does not imply a stock position.
Operating Temp. Range			-20 ~ +70, -40 ~ +85			˚C	
Frequency Stability (Overall, 20 Years)					±4.6	ppm	Including calibration @ 25°C, supply voltage V_{DD}±5% , load ±10%, reflow soldering, 20 years aging and frequency stability over temperature.
Frequency Stability Vs Temp. Range					±0.1 (-20~+70°C) ±0.2 (-40~+85°C)	ppm	Ref. to (FMAX+Fmin)/2
Holdover Stability					0.37	ppm	Including 24hours aging , supply voltage V_{DD}±5% and frequency stability over temperature.
Supply Current	CMOS				8	mA	
	Clipped Sine Wave				5	mA	
Output Level	CMOS	Output High (Logic "1")	90%V_{DD}			V	
		Output Low (Logic "0")			10%V_{DD}	V	
		Duty	45		55	%	
	Clipped Sine Wave		0.8			Vp-p	
Load	CMOS			15pF			
	Clipped Sine Wave			10 KΩ // 10pF			
Phase Noise @ 10MHz	@ 100Hz				-125	dBc/Hz	Please add filter capacitor as suggested, and 33nF at pin7
	@ 1kHz				-145	dBc/Hz	
	@ 10kHz				-154	dBc/Hz	
G Sensitivity				0.5	1.0	ppb/G	Each Axis, 20-2000Hz
Start Time					10	mSec	
Storage Temp. Range			-55		+125	˚C	
VCTCXO	Control Voltage Range		0.5		2.5	V	
	Pulling Range		±5.0			ppm	
	Vc Input Impedance		100			kΩ	

Note: not all combination of options are available. Other specifications may be available upon request.
Specifications subject to change without notice.

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